IBD Plexus

Proposal Submission and Project Selection Guideline

CROHN'S & COLITIS FOUNDATION

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1. Overview

IBD Plexus, named for its complex network of parts, was founded with a mission to advance science, accelerate progress towards precision medicine and improve the care of patients living with inflammatory bowel diseases. The goal of IBD Plexus is to accelerate research in high impact areas through the creation of a research exchange system designed to nurture partnerships among IBD stakeholders (patients, caregivers, providers, academic and industry researchers) to increase collaboration and sharing of data. The national-scale, cloud-based platform integrates clinical, patient-reported, genetic and molecular data from diverse research initiatives, real world clinical care settings and patients' experiences. IBD Plexus provides academic and industry researchers with access to research-ready datasets and analytical tools to more rapidly perform activities that promise to speed treatment development, optimize existing therapies through creation of biomarkers and diagnostics and improve health outcomes.

It is important that IBD Plexus is a source of high scientific integrity and that publications and IP generated from IBD Plexus resources are high impact. To ensure scientific merit and that the use of biosamples are optimized across the researcher community the following guidelines have been developed to outline the proposal submission and project selection process.

Premier Founding Members and Contributing Sites can submit requests to gain access to the following:

Clinical, Patient-reported data and molecular data: Patient-level raw data collected from various sources including:

- Longitudinal, clinical patient-level data from a subset of the IBD Plexus cohorts
- Patient-reported data from a subset of the IBD plexus cohorts (SPARC IBD, IBD Qorus, IBD Partners)
- Genetic and molecular ('omics) patient-level data generated by researcher or derived by the Foundation

Biosamples: Biosamples from a sub-set of the IBD Plexus cohorts (SPARC IBD, RISK)

Private Collection: Development of a specific protocol to capture additional data and /or biosamples from IBD Plexus sites.

Chart review: Review of clinical notes extracted from medical records

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The rigor of the scientific review process is dependent on the request type. The below chart outlines the request requirements regarding whether or not a proposal is necessary for the different request types:

Request Type	Member Type	Requires Proposal Approval?
Clinical, patient-reported data	Premier Founding Industry Members	No
	Contributing Sites	No - Site-specific data Yes - other sites / cohorts data
Molecular data	Premier Founding Industry Members	No
	Contributing Sites	No - Site-specific data Yes - other sites / cohorts data
Biosamples	Premier Founding Industry Members	Yes - capped at 6
	Contributing Sites	Yes - capped at 6
Private Collection (i.e. surveys)	Premier Founding Industry Members	Yes
	Contributing Sites	Yes
Clinical Trial Site Identification	Premier Founding Industry Members	No
	Contributing Sites	No

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Cycle 2021	Proposal Central Opens	Proposals due
Cycle 1	01/01/2021	01/31/2021
Cycle 2	03/01/2021	03/31/2021
Cycle 3	05/01/2021	05/31/2021
Cycle 4	07/01/2021	07/31/2021
Cycle 5	09/01/2021	09/30/2021
Cycle 6	11/01/2021	11/31/2021

Note: For contributing site proposals, the site principal investigator or co-investigator from a contributing site must be the lead researcher on the submission.

2. IBD Plexus Project Selection Committee

The Project Selection Committee is charged with evaluating proposals. It makes decisions to approve, revise and resubmit, or deny Proposals, and will monitor use of resources, help regulate and prevent duplicative projects, and encourage collaboration, where appropriate.
The Project Selection Committee is comprised of:

- A subset of researchers from each cohort
- A biostatistician
- Independent researchers (not affiliated with any cohort)
- Patient(s) and caregiver(s)

Project Selection Committee members have 3-year term limits and are subject to the Foundation's conflict of interest policies. Members are required to remove themselves from any reviews and/or discussion that could pose a conflict of interest as defined by Foundation's Peer Review Integrity Policy.

3. Proposal Review criteria

Proposals will be reviewed against criteria including:

- Impact / priority
- Significance to IBD
- Innovation
- Approach
- Investigator
- Quality of research environment
- Value to IBD Plexus and the broader scientific community

Members of the IBD Plexus Project Selection committee will evaluate and rate each proposal using the scorecard in Appendix 1.

IBD Plexus is intended to facilitate research that will accelerate precision medicine, enable the development of efficacious treatments, and ultimately identify cures for patients leaving with IBD. In this vein, proposals requesting resources explicitly for non-permitted commercial uses will be denied.

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Examples of non-permitted commercial uses include product promotion, marketing, targeting segments of the physician IBD landscape to understand prescribing patterns or re-identifying patients.

a. Biosamples request

For proposals requesting biosamples access, additional criteria will be evaluated:

- Rational of biosamples request
- Size of request for biosamples
- Laboratory proposed for the biosamples analysis*

*IBD Plexus has selected central reference laboratories to ensure standardization across technology, Standards of Procedures and assays, and eliminate variables that affect test results and uniformity. Each central reference lab is a leading expert in one or more of the following:

- Microbiome
- Genomics
- Transcriptomics
- Metabolomics
- Proteomic

It is highly recommended that researchers consult with the IBD Plexus Research Team prior to submit a proposal. All proposals requesting biosamples will be vetted by an independent biostatistician to ensure the proposed biosamples size is sufficiently powered (and not excessive) to validate the proposal's hypothesis.

b. IRB approval

If a proposal includes a request for the use of a Limited Data Set, as defined by HIPAA (including in connection with a proposal requesting biosamples), researcher must obtain IRB approval prior to receiving access to the Limited Data Set

c. Conflict of Interest

For requests with perceived conflict of interest (as determined by the Project Selection Committee), such as a submission perceived to be requesting the collection of new data that might be used to advance its positioning in the marketplace, the Proposal must be reviewed by the Foundation Ethics Committee to determine if there is a conflict. If the Project Selection Committee disagrees with the Ethics committee recommendation, member can initiate an appeal to the Foundation Oversight Committee. The Foundation Oversight committee will then make the final decision.

d. Approvals / Denials / Appeal

Pending Approval with Stipulations:
The Project Selection Committee may suggest revisions to and re-submission of a proposal pending approval. The Researcher has the option to revise and re-submit the Proposal through an expedited process if the recommendations suggested by the Project Selection Committee have been implemented.

Denial/Appeal:

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In the event that a proposal is denied (e.g., due to insufficient evidence for a hypothesis, unclear description or determination that an idea is scientifically premature), the Company may submit a letter of appeal to the Appeals Committee, which is composed of an external expert/researcher, the National Scientific Advisory Committee (NSAC) Chair, and Foundation Director of Translational Research. The Appeals Committee will review the Proposal to determine the appropriateness of a new review.

4. Targeted / Non-Targeted Data

At the time of proposal review, the Project Selection Committee will determine whether it approves the Proposal as a Targeted Project or Non-Targeted Project. For any disputes of proposal classification (e.g. Targeted Project vs. Non-Targeted Project), researcher may make an appeal to the Project Selection Committee for re-classification no later than forty-five (45) days before the expiration of the Non-Targeted Exclusivity Period or Targeted Exclusivity Period, as applicable.

Two exclusivity periods have been defined based on data type. Non-targeted data derived from biosamples is not considered knowledge while targeted derived data can potentially contain competitive knowledge due to the insights that could be gleaned from the data. Please see definitions below

- **Non-targeted data**
  Definition: Generated data that is not directed at a particular hypothesis but rather is a preliminary exploration of the data. (e.g., RNA sequencing)
  - Generation of non-targeted data, whenever possible, should be generated at an IBD Plexus designated core laboratory. In select cases for which an IBD Plexus designated core laboratory does not have the capability to generate non-targeted data, IBD Plexus will approve receipt by recipient of samples so that recipient may generate non-targeted data.
  - If the project generates non-targeted data, the researcher must deposit data generated into the IBD Plexus platform as per whichever milestone below is reached first
    - 180 days (6 months) after recipient's last receipt of resources
    - Acceptance of a project manuscript for publication

- **Targeted data**
  Definition: *a) Data that has been generated to answer a focused hypothesis (e.g., association between three thiopurine metabolites and mucosal healing in patients with Crohn's Disease) b) Data that has been generated using a non-commercially available assay*
  - If the project generates targeted data the researcher must deposit data generated into the IBD Plexus solution as per whichever milestone below is reached first
    - 540 days (18 months) after recipient's last receipt of resources
    - Acceptance of a project manuscript for publication
    - Recipient's filing of a patent application
  - Researchers can submit a request for an extension as long as there has been adequate progress of the research toward publication or patent filing as determined by the Project Selection Committee.
    - Extension requests must be submitted by month 14
    - Extensions will be capped at one additional year of data exclusivity

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5. Duplicate or overlapping proposals

For proposals found to have duplicate or overlapping project scopes, the Project Selection Committee will assess the degree of overlap with existing projects as part of the review process. If more than one researcher is requesting the same biosample(s) to generate the same data, the Foundation will notify the researchers of the overlapping request. If there is substantial overlap, an effort will be made to encourage collaboration without revealing the identity of the researchers.

1. If the researchers agree to collaborate, then they will jointly share ownership of and exclusivity rights in and to the project analyses and biosample-derived data. The researchers will agree on how to allocate costs associated with the Proposal between them.
2. If the researchers do not agree to collaborate, the Project Selection Committee will rank the overlapping proposals based on which proposal is based on the best science and will approve one researcher's proposal and deny the other Researcher's proposal.
3. In cases where the scientific merit of the overlapping proposals is equal (or similar), the Foundation will assess on a case-by-case basis.

6. Process for submitting Proposal:

The application should be submitted to proposal central at: https://proposalcentral.altum.com

To submit a proposal follow the steps below:

1. Establish a clinical or translational research question that can be answered within IBD Plexus resources.

2. Log into the IBD Plexus Research Portal https://www.ibdplexus.org/ and use Explore Cohorts in Population Analyzer to assess data and/or biosamples availabilities and refine your hypothesis leveraging IBD Plexus Pre-to Research tools.

3. Review the Approved Project list to determine if a similar hypothesis has been or is currently being studied.

4. Arrange a free one-on-one consultation with the IBD Plexus Research team or one of the PI of an IBD Plexus cohort to help develop your proposal. Researchers are encouraged to seek support for proposal development and submission from one of the cohort's investigators, which may include a one-on-one consultation to discuss the Proposal in advance of the submission. The purpose of the consultation is to guide understanding of the IBD Plexus data and biosamples in order to submit a well-informed proposal to streamline the submission/review process.

5. To submit your proposal, members must have or create an account in Proposal Central.
    - If you are a first-time user, navigate to https://proposalcentral.altum.com and register by clicking on "First time user." This will generate a confirmation number which will be emailed to the email connected to this account. Confirm your registration by submitting this number when prompted.
    - If you have already registered to Proposal Central, enter your username and password and submit.

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6. After login in, click on “Grant opportunities” on the far right of the page and select “Crohn’s & Colitis Foundation” under ‘Filter by Grant Maker” drop down menu on the upper left of the page.

7. Locate “IBD Plexus” announcement and click “Apply now" to submit a Letter of Intent (LOI). The Letter of Intent is mandatory for all new applications. Resubmitted applications do not require a letter of intent**.**
The LOI will include:
    - Project title (do not use abbreviation unless absolutely necessary)
    - Member Type (select the appropriate answer)
    - Request type (select the type of request) and enter additional request type in the free text field if more than one type of request if submitted
    - Challenge Area (select the appropriate answer)
    - Applicant/PI information
    - Institution or Company information

Abstract

- **Scientific Summary of the Project**
  The Scientific Summary should provide a clear, concise overview of the proposed work, including the background, objective, or hypothesis and its supporting rationale; specific aims of the study for the general scientific audience
- **IBD Plexus Usage Rational**
  The rational should include why your project should use IBD Plexus and how it will further both research and IBD Plexus mission

8. Once the LOI is submitted and approved, you will receive an email notification from Proposal Central to submit a full proposal.

6. The following documents will need to be attached to your proposal within *Proposal Central*:
    - Proposal title
    - Challenge area. Select the priority area that is addressed by the proposal
    - A cover letter stating the overall purpose and rationale of the proposed research study.
    - NIH format biosketches / CV of the PI
    - Relevance of the proposal to IBD Plexus’ mission. Provide a rational for the use of IBD Plexus resource and how this study will further both IBD Plexus and Foundation’s mission
    - A study proposal, not to exceed **6 single-spaced pages,** that provides the
        - hypothesis,
        - specific aims,

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- rationale,
- Design
- For proposals requesting biosamples access, additional information should be included:
    - Rational of biosamples request
    - Number of biosamples requested and supporting rational
    - Laboratory proposed for the biosamples analysis*

- Analysis plan for the data and/or biosamples
- General statement of project impact in IBD
    - Lay summary. Lay patients participate as voting members of the IBD Plexus Project Selection Committee. The lay summary should be a clear, concise overview in simplified language, appropriate for non-scientific reviewers. You need to provide enough essential information that the lay reviewers are able to evaluate your application. The lay summary should include the following information:
        - What question will this project attempt to answer?
        - Why is this question important to IBD?
        - What is the study design?
        - How do the hypothesis and specific aims fit with IBD Plexus and Foundation's priorities?
        - Will this research project, if successful, further IBD Plexus' mission to accelerate progress toward precision medicine and improve quality of like for IBD patients? If so, how will this project do so?

*IBD Plexus has selected central reference laboratories to ensure standardization across technology, Standards of Procedures and assays, and eliminate variables that affect test results and uniformity. Each central reference lab is a leading expert in one or more of the following:

- Microbiome
- Genomics
- Transcriptomics
- Metabolomics
- Proteomics

It is highly recommended that researchers consult with the IBD Plexus Research Team prior to submit a proposal. All proposals requesting biosamples will be vetted by an independent biostatistician to ensure the proposed biosamples size is sufficiently powered (and not excessive) to validate the proposal's hypothesis.

Note: Additional details may be requested from the investigators based on the initial review by the IBD Plexus Project Selection Committee.

Note: Titles and lay summaries of your Letter of Intent and your full proposal will become public information and will be shared in a public domain. Therefore, please do not post a title or lay summary that includes proprietary/confidential information.

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7. Proposal cycles

Proposals will be reviewed by the Project Selection Committee. All Proposals are reviewed six (6) times annually for Founding Premiere Members and three (3) times annually for Contributing Sites.

The Project Selection Committee will review and render a decision (either approve, revise and resubmit, or deny) within eight (8) weeks of the date that the Project Selection Committee reviewed the Proposal. Note: For Premiere Founding Members IBD Plexus Data shall be provided upon request (requires only an administrative review) and does not need to go through a formal project selection committee review.

8. Request fulfillment

Once a proposal is approved, researchers will log into https://www.ibdplexus.org/ to submit a data fulfillment request to the Foundation. Requested IBD Plexus data will be provided through one of three mechanisms based on the researcher's preference:

- Encrypted hard drive
- Electronic transfer protocols, such as hypertext transfer protocol secure, or HTTPS, and secure file transfer protocol, or SFTP
- High performance computing: Ability to provision high performance computing private research work area in the IBD Plexus cloud

Researchers must abide by the IBD Plexus data use agreement which states that data cannot be used for non-permitted commercial use, or the use of data for product promotion, marketing, targeting segments of the physician IBD landscape to understand prescribing patterns or re-identifying patients, or any other uses that the Foundation Ethics Committee deems as unethical.

Biosamples fulfillment

The Foundation will provide requested biosamples along with a Material Transfer Agreement ("**MTA**"), which the researchers negotiate and agree, setting forth the terms and conditions governing operational matters including shipment of biosamples, applicable additional costs and protocol for conducting the proposed analyses.

If biosamples are sent to one of the Foundation's IBD Plexus central reference labs, the Foundation will provide the researcher with the biosample-derived data relating to such biosamples. If biosamples are sent to a non-Foundation central reference lab, the researcher must provide the Foundation with the biosample-derived data relating to such biosamples.

9. Publication requirements

a. Publication Review

- Within thirty (30) days after creation of the final draft of any abstract, poster or paper analyzing or otherwise utilizing resources and prior to its submission for publication

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researcher must provide the IBD Plexus Publication Review Committee with a copy of the abstract or manuscript when applicable. Researcher should use best efforts to seek publication of any project paper in accordance with generally accepted scientific standards.

- Project papers will be reviewed by subset of the Publication Review Committee on an ad-hoc basis, provided that such review shall not exceed thirty (30) days from receipt of the project paper by such Committee. If there are no concerns with publication, the researcher may move forward with publication efforts; however, if review of manuscript indicates that issues need to be addressed, the project paper will be reviewed by the entire Publication Review Committee, provided that such review shall not exceed sixty days from receipt of the Project Paper by such Committee. Reasons for rejection or additional review include, but are not limited to, the following:

    - Manuscript contains one or more egregious methodological errors
    - Manuscript generates major reviewer concerns regarding methodology, findings and/or conclusions that are found to misinterpret the data and/or cohorts
    - Manuscript is a market research analysis (e.g., prescribing patterns)
    - Manuscript falls outside the scope of the approved Project.

1. In the event that Researcher's project paper has been rejected, the Researcher shall be given the opportunity to revise the project paper and resubmit it for review.

2. The Foundation may make any project paper available through IBD Plexus after its publication or after the first anniversary of its completion, whichever is earlier, provided that it is permissible by the publishing journal and doing so does not compromise the publishability of any project paper.

b. Publication acknowledgment

Acknowledgment of IBD Plexus – All Project Abstracts, Posters and Papers must include acknowledgement of the Foundation, e.g., "The results published here are, in whole or part, based on data or biosamples obtained from the IBD Plexus program of the Crohn's & Colitis Foundation".

Appendix 1

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IBD Plexus Project Selection Committee Review Criteria

Please provide prose or bullet points of the strengths and weaknesses, and use the scale, referenced at the bottom of this page, to score from 1-9 (1 = exceptional, 9 = poor) for each section.

REVIEWER:	
INVESTIGATOR:	
PROPOSAL TITLE:	
DATE:	

Score ☐ **OVERALL IMPACT/PRIORITY:** *Give an overall score based on all of the review criteria and overall scientific merit. If Academic proposal, particularly consider the likelihood that the proposal will position the applicant for additional funding (e.g., NIH grant)*

Score ☐ **SIGNIFICANCE:** *Does the proposal address a significant knowledge gap in basic or clinical understanding of IBD or related gastrointestinal diseases? Is the proposal likely to lead to significant gains?*

Score ☐ **INNOVATION:** *Does the application propose novel concepts, approaches or methodologies, aimed at improved bench to bedside research in IBD and driving progress to precision medicine or improving the overall health of patients with IBD? Assess the translatability of the proposal.*

Score ☐ **APPROACH:** *Include a comment on whether the proposed work is appropriate in scope and the methods are feasible.*

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Score ☐ **INVESTIGATOR(S):** *Is this an experienced IBD investigator with a track record of high quality productivity and collaboration in the field? If a junior investigator, do they have a mentor/mentoring team with the appropriate qualifications to supervise the work?* (NIH Biosketch to be reviewed for Academic researchers and NIH Biosketch or work experience section to be reviewed for Industry researchers)

Score ☐ **ENVIRONMENT:** *Comment on the IBD Plexus resources that will be used: data and biosamples. Is the investigator in the right environment to make optimal use of the IBD Plexus resources?*

Score ☐ **VALUE TO THE NETWORK:** *Will this project add new data into the IBD Plexus platform? If so, are the new resources created following IBD Plexus standards and useful to the research community?*

Is the researcher requesting biosamples? Yes ☐ No ☐

Projects which will utilize biosamples will need to receive an outstanding impact score, and have a high likelihood of success. Careful attention will be paid to the amount of the biosample requested.

If yes, please fill out the additional review criteria

Score ☐ **BIOSAMPLE USAGE:** *Does the rationale of the study justify use of the requested samples? Is the number of samples requested reasonable and sufficiently powered to achieve statistical significance? Type of biosample (tissue, blood, stool)*

Score ☐ **LAB VENDOR:** *Is the investigator using IBD Plexus reference lab? If not, is the lab that the investigator is proposing to use for molecular analysis considered high quality? Is the lab willing to provide the necessary metadata back to IBD Plexus and also provide the data back in a format that can be easily migrated into IBD Plexus? Does this proposal merit use of an outside lab?*

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Additional Scoring Notes:

- The project selection committee will subtract 1 point from the overall score for industry proposals.
- Proposals that score 3 or lower (depending of resources available and applicant pool), will be approved.

Please use the following 1-9 scoring scale

Impact	Score	Descriptor	Additional Guidance on Strengths/Weaknesses
High	1	Exceptional	Exceptionally strong with essentially no weaknesses
	2	Outstanding	Extremely strong with negligible weaknesses
	3	Excellent	Very strong with only some minor weaknesses
Medium	4	Very Good	Strong but with numerous minor weaknesses
	5	Good	Strong but with at least one moderate weakness
	6	Satisfactory	Some strengths but also some moderate weaknesses
Low	7	Fair	Some strengths but with at least one major weakness
	8	Marginal	A few strengths and a few major weaknesses
	9	Poor	Very few strengths and numerous major weaknesses

Minor Weakness: An easily addressable weakness that does not substantially lessen impact
Moderate Weakness: A weakness that lessens impact
Major Weakness: A weakness that severely limits impact

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